Msc Science Education and Communication

MATHerials:
an action-based embodied design for the sine graph

Internship Product Development
Product development report

Jagna Stella Hendriks

Supervisors:

Dr. Anna Shvarts
Freudenthal Institute Utrecht University

Rosa Alberto
HU lectorate mathematic and analytic faculties for professionals

June 26, 2026

Contents

1 Introduction

Mathematics and numeracy are increasingly important in professional and personal life (Foley et al., 2017). Acquiring mathematical and numerical skills are mainly the domain of (formal) education. Many factors impact the process of acquiring these skills, such as math anxiety and teaching methods.

Embodied cognition theory may shed some light on how to reduce math anxiety and foster conceptual understanding among students. Embodied cognition theory is a theory within cognitive science which assumes that the interaction with the environment is fundamental for cognitive processes. Working from this perspective educational material would need to be designed differently to accomodate learning.

For more insight in how embodied cognition theory can be applied within education, research is being conducted on embodied designs. Embodied design is a collection of different approaches and design principles to make educational materials based on embodied cognition theory. One group of researchers, who are researching embodied designs, is MATHerials. Some researchers within this group are designing, implementing and studying action-based embodied designs which is a subcategory of embodied designs. Such designs are mainly meant as tools to be used by teachers in secondary schools to enhance pupils' understanding of mathematical concepts such as proportions, angles, parabolas and the sine function. A particular interest at the moment is the development of physical rather than digital action-based embodied designs for use in secondary schools. For already existing digital designs this poses the challenge of translating them into physical designs.

The product will be a translation of a part of the digital action-based embodied design concerning the sine graph. To understand what the product should be, we will analyse the problem it needs to solve and more in section 2. This information will then inform the product development as set out in section 3. To see whether the product meets expectations, the product will go through a small evaluation which will be discussed in section 4.

2 Analyses

In this section we will give the theoretical basis for the product by discussing the problem, sender, desired effects, target audience, concept and practice and approach analyses. This section will be concluded with a summary of the most important points which need to be taken into account during the development of the product. A more complete theoretical background for the product can be found in the theory report.

2.1 Problem analysis

The term math anxiety is commonly used to express the fear, tension and apprehension people feel towards or when engaging with mathematics. Luttenberger et al., 2018, p. 312 states that across 34 participating countries " *[...] 59% of the 15- to 16-year-old students reported that they often worry that math classes will be difficult for them; [...]*". This together with other statistics mentioned in Luttenberger et al., 2018, Foley et al., 2017 and Ramirez et al., 2018 shows that math anxiety is a global phenomenon which impacts a significant portion of the population. Results from different experiments and surveys on math anxiety point towards a negative correlation between math anxiety and performance. A suggestion to reduce math anxiety among pupils is posed in Luttenberger et al., 2018, p. 319, namely that "*Teachers may choose instructional strategies that enhance students' interest and motivation [...]*" and "*Similar advice involves the use of hands-on devices and manipulatives in learning.*" With this advice embodied designs give rise to a possibility to reduce math anxiety among students. Since embodied design is a collection of different approaches to make educational materials based on embodied cognition theory, we will shortly discuss embodied cognition theory.

Embodied cognition theory is a relatively young discipline which arose as a movement against mainstream cognitive science theories such as cognitivism and behaviourism (Núñez et al., 1999, Shapiro and Stolz, 2019). It encapsulates many different hypotheses which range from conservative to radical (Boonstra et al., 2023, Farina, 2020, Núñez et al., 1999). The range of conservative and radical hypotheses share a common assumption, namely that the interaction with the environment is fundamental for cognitive processes (Boonstra et al., 2023).

Boonstra et al., 2023 states that the process of understanding a mathematical concept, also refered to by conceptual understanding, is a key component of mathematical thinking. As described in Gravemeijer et al., 2016 current mathematics education based on realistic mathematics education (RME) is not yet equipped to facilitate the development of more conceptual knowledge of mathematics. Perspectives on learning from embodied cognition theory could provide insights into which tools for mathematics education can improve the development of more conceptual knowledge.

Both RME and embodied cognition strive to facilitate conceptual learning of mathematics and to create relevance for students. As indicated earlier creating relevance for students has a potential to reduce math anxiety. As a continuation of RME and embodied cognition, MATHerials produces and conducts research into educational material based on embodied cognition theory, also known as embodied designs. Quite some designs have already been produced and researched by MATHerials. Many categories of embodied design exist, such as action-based embodied design. This type of embodied design will be the focus of the product. MATHerials has some digital action-based embodied designs introducing some concepts, such as proportions, angles, parabolas and the sine graph. However, Rosa Alberto (personal communication, February, 2026) has noticed that although digital designs give much freedom from a design perspective, they also have some drawbacks. A drawback of the digital action-based embodied designs is the need for touch-screen. Tablets would be necessary to use the digital action-based embodied designs effectively in classrooms, which is not feasible for every school. Therefore, the challenge to translate the digital designs into physical designs arose. Here we will focus on translating a part of the digital design for the sine graph made by MATHerials into a similar physical design with certain requirements. The requirements will be adressed in subsection 2.2, and the digital action-based embodied design for the sine graph will be illustrated in subsection 2.5.

2.2 Desired effects analysis

The product will need to be an action-based embodied design. Action-based embodied designs are designs which pose a motor problem with (continuous) performance feedback (Alberto et al., 2022). A motor problem is a problem which can be solved by a certain movement. Users are meant to struggle to produce the movement required to solve the motor problem. This struggle to produce this movement will help the user to actively engage in seeking a strategy for the solution. This strategy can later be verbalised to promote translating the movement into a mathematical insight. Action-based embodied designs consist of two phases, namely a qualitative and a quantative stage (Alberto et al., 2022). Each stage has an acting and a reflective step. In the acting step students explore the problem, and in the reflection step students are encouraged to describe their experience and solution in a culturally normative way. The difference between the qualitative stage and the quantative stage is that in the quantative stage the same motor problem has cultural artifacts such as grids to shift students' perception and reasoning.

Eventually the aim of the design will be to be used as (extra) lesson material in classes. A couple of goals which work towards this larger aim are: an affordable product and a product which is easy to incorporate into existing classes. An affordable product will ensure that schools have the resources to purchase the product in quantities which enable the product to be used efficiently in classes. A product which is easy to integrate into existing classes will encourage teachers and schools to try this product (Rosa Alberto, personal communication, March, 2026). As mentioned by Rosa Alberto (personal communication, March, 2026) and Annemiek van Leendert (personal communication, 8 April, 2026) - a researcher of mathematics education for visually impaired students and educator - a way to make the product easy to incorporate into lessons is to make student worksheets for the product and a teacher guide on how and when the product could be used.

Moreover, the product is meant to foster a better understanding of mathematics and less math anxiety among students of secondary schools. More specifically the product is meant to facilitate conceptual understanding of the construction of a sine graph using a unit circle. de Freitas et al., 2018 found that participants deepened their embodied understanding of a geometric figure during the teaching experiment. In a similar way as in de Freitas et al., 2018 the product should deepen the embodied understanding of the construction of a sine graph using a unit circle. de Freitas et al., 2018 also found that the teaching experiment encouraged affective communication and sympathy. The product should also encourage affective communication and sympathy, primarily to mitigate math anxiety.

2.3 Sender analysis

The internship takes place within the Hogeschool Utrecht (HU) lectorate mathematical and analytical faculties for professionals, which focuses on creating evidence-based practices with educators to strengthen the mathematical and analytical faculties of their pupils. MATHerials is part of the effort of this lectorate to research and communicate possibilities for improvement of mathematics education. The lectorate mathematical and analytical faculties communicates in a number of ways, such as organising focus groups, participating in and organizing conferences, maintaining different informational websites and posting on LinkedIn.

A priority of this lectorate of the HU is to prepare future professionals and citizens to be able to actively participate in society as can be seen in HU, 2026. As mentioned in section 1 numeracy is becoming more important for both professionals and citizens in daily life. Part of this priority is improving education concerning mathematical and numerical skills. Improving education can be approached in different ways. For example for students it is important to enhance conceptual understanding and decrease math anxiety such that they can use numerical skills with confidence. Furthermore, for educators it is important to know how to facilitate conceptualisation among students, and how to reduce math anxiety. The development of educational material itself is also of importance as this informs future educational practices.

2.4 Target audience analysis

The main target audiences for the educational product are pupils and teachers. Moreover, the product will also serve as an example of a physical action-based embodied design for (educational) designers.

For the product we will be focussing on fourth year students in secondary school. In this year the sine function is usually introduced, as can be seen in subsection 2.5 where the sine function as explained in Bakker et al., 2014 is introduced. Attitudes towards mathematics (education) may vary between pupils. However, as mentioned in subsection 2.1 more than half of 15- to 16-year-old students mention feelings of anxiety towards mathematics (Luttenberger et al., 2018).

According to Boonstra et al., 2023 teachers struggle to facilitate conceptual understanding among students. The focus in Boonstra et al., 2023 lies on geometry teachers who often lack sufficient knowledge and struggle with feelings of insecurity with regards to teaching geometry. Therefore, the product also focuses on mathematics teachers and how to support teachers in encouraging conceptualisation among pupils.

To encourage more similar learning experiences in schools, educational designers need to see the possible kinds of designs. The product in combination with a communication strategy, such as a flyer about the design principles of an action-based embodied design or an instruction manual, will provide an example for educational designers on how to make (action-based) embodied designs for mathematics education.

2.5 Concept analysis

The product will mainly be concerned with conveying the connection between a unit cirle and the sine graph through a physical action-based embodied design. Specifically, the product will focus on the x-coordinate of the sine graph which is connected to the circumference of a unit circle. Furthermore, the product will be accompanied by a small and limited list of questions which can be answered by using the product. Moreover, an indication of when the product could be used in classes will be given. However, providing worksheets for students and a full teacher guide is not feasible during this internship.

In this section we will shortly discuss the connection between a unit cirle and the sine graph as mentioned in a Havo 4 mathematics book (Bakker et al., 2014) and discuss an example of an action-based embodied design for the sine graph. The concept map in Figure 11, which can be found in section 7, shows the important aspects of the connection between the unit circle and the sine graph based on the following subsections.

2.5.1 Unit circle and sine graph

In Bakker et al., 2014 the concepts radials and the unit circle are introduced first after which the sine graph is discussed. For an intuitive approach of the sine function, it is first introduced using a figure similar to Figure 1. Using such a figure, it is explained that when moving a point over the unit circle the height can be determined by $\sin(\theta)$ where θ gives an indication of the distance travelled over the circle (part of the circumference).

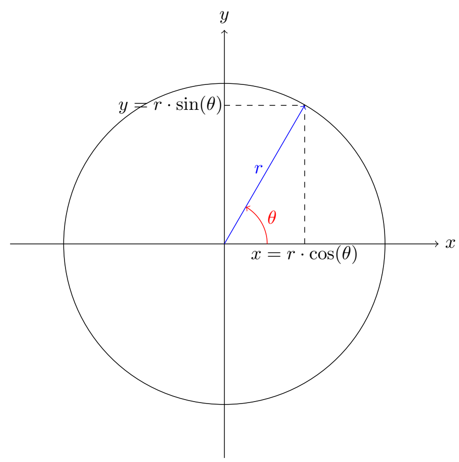


Figure 1: A circle with the sine and cosine. For a unit circle one has $r = 1$.

Using some intermediary figures and exercises, the theory is lead in the direction of the sine graph. In Bakker et al., 2014, chapter 8 the sine function is further explained using a figure similar to Figure 2 and a few facts about the sine graph are listed. Some of the facts mentioned in Bakker et al., 2014, p. 202 are:

- The midline of the graph of this function is the line $y = 0$ (x-axis).
- The amplitude is 1.
- The sine function is periodic and its period is 2π.
- At the point $(0, 0)$ the graph increases crossing the midline. Every point at which the graph increases when crossing the midline, is called a starting point.

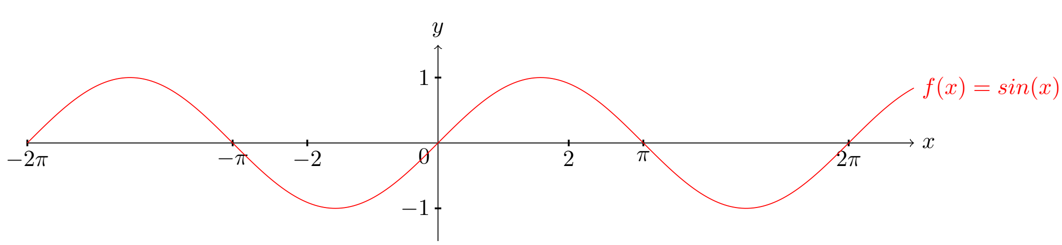


Figure 2: The graph of $f(x) = sin(x)$.

2.5.2 Action-based embodied design for the sine graph

To illustrate the principles of action-based embodied design, we will discuss the digital design for the sine graph as explained by Shvarts and van Helden, 2021. This action-based embodied design focuses on the connection between the sine graph and a unit circle. This design consists of four parts as can be seen in Figure 3.

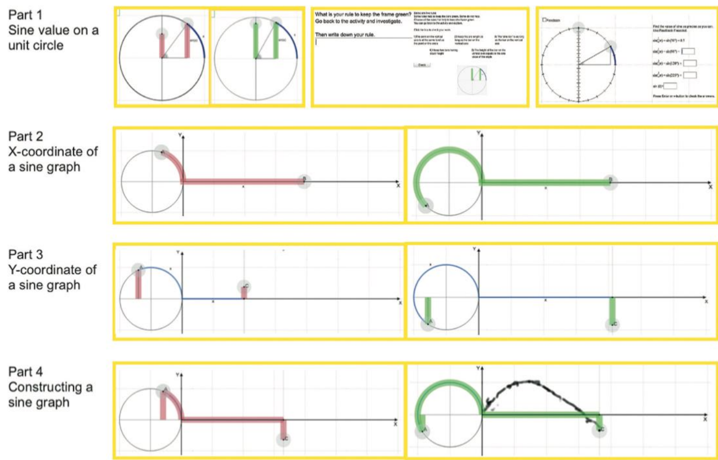


Figure 3: This figure is taken from Shvarts and van Helden, 2021. It depicts the tasks in the learning sequence of the digital action-based embodied design for the sine graph.

Every part of the design has an acting and a reflective step. In the active step students are expected to interact with the motor problem. In the reflective step students are expected to reflect on their solution startegy in order to gain more understanding of the mathematical concept behind it.

In every part of this design students need to keep the feedback green in the active step. In part one the feedback becomes green when the height on the y-axis is the horizontal projection of the cursor on the unit circle. This is to familiarise students with the value of the sine function on the unit circle. In part two the feedback becomes green when the length on the x-axis is the same as the circumference of the unit circle up to the cursor. In this part students enact the x-coordinate of the sine graph in connection to the unit circle. In part three the length on the x-axis is automatically changed with respect to the circumference travelled up to the cursor. In this part the feedback becomes green when the height matches the height of the cursor in the unit circle. This is to familiarise students with the y-coordinate of the sine graph as viewed from the unit circle. Part four combines parts two and three to construct the sine graph from the unit circle.

The latest version of these tasks can be found on https://embodieddesign.sites.uu.nl/distant-learning/.

The physical design of part two is challenging in that circular motion needs to be translated to linear motion for a feedback mechanism. This challenge will require some time to work out. Therefore, to make the product feasible within the time-frame of the internship only part two of this digital action-based embodied design will be translated into a physical product.

2.6 Practice and approach analysis

This product mainly centers around designing a physical teaching tool. Figure 10 shows possible techniques, materials and components which could be used to produce a physical design.

A design process is crucial for designing products as expressed by the following quote from Dubberly, 2004: "*If we wish to improve our products, we must improve our processes; we must continually redesign not just our products but also the way we design.*" Dubberly, 2004 offers many different processes for designing. Two interesting design processes are: Human-centered design processes for interactive systems (Dubberly, 2004, p. 75) see Figure 4, and the iconic model of the design process by Mihajlo D. Mesarovic (Dubberly, 2004, p. 121) see Figure 5.

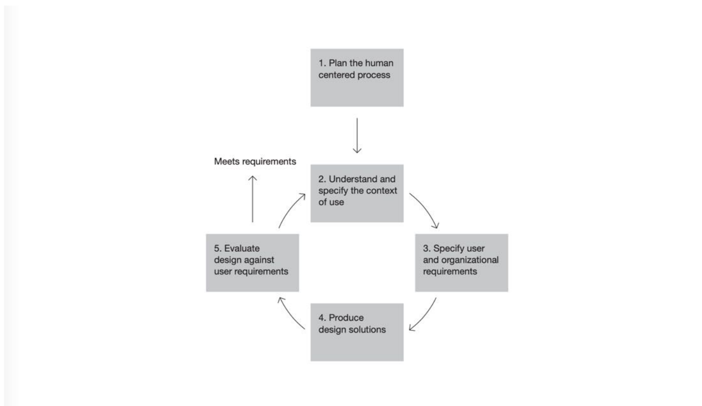


Figure 4: This figure is taken from Dubberly, 2004, p. 75. It depicts the Human-centered design processes for interactive systems.

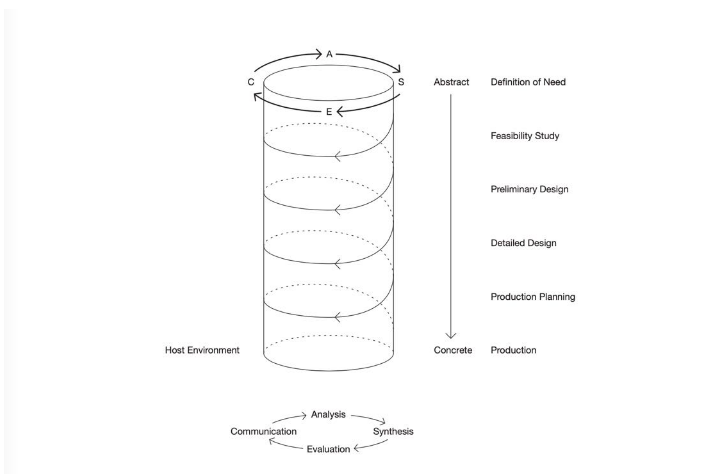


Figure 5: This figure is taken from Dubberly, 2004, p. 121. It depicts the iconic model of the design process by Mihajlo D. Mesarovic.

In order to produce a physical product the digital design needs to be translated into a physical design. Based on the design processes mentioned above, we will formulate a short design process here. The first step of the process will be coming up with multiple ideas and creating prototypes. By evaluating these first prototypes a more detailed physical design can be formulated. Once a new design has emerged, this in turn will become a prototype and be evaluated. Due to the limited amount of time this process will be repeated two to three times, with different evaluations.

Part of the design process is evaluating a prototype of the product to gain feedback and improve the next prototype. The evaluations will take the form of interviews. During these interviews, we will shortly introduce a prototype and let the participants explore the motor problem the prototype poses.

While exploring the motor problem we will ask questions, such as what would be the (mathematical) rule behind this exercise. Afterwards we will ask feedback about the prototype, such as what they thought of the feedback mechanisms used by the prototype, what they learned and whether they enjoyed the experience.

2.7 Summary

As discussed in subsection 2.2 the product should be an action-based embodied design which is affordable and easy to integrate into existing lesson materials. The importance for the physical action-based embodied design lies in creating a feedback mechanism and making sure the solution to the motor problem is not pre-determined. Producing an affordable product can be achieved by using cheap materials and techniques to build the product. Producing a product which is easy to integrate into existing lesson material can be achieved by making a preliminary worksheet and giving an indication of when the product could be used in existing lessonplans.

The product will be a translation of the digital action-based embodied design as described in subsubsection 2.5.2. Specifically part two (the x-coordinate of the sine graph) of the digital action-based embodied design will be translated into a physical product due to time constraints of this internship.

Creating a physical design will require a design process. As described in subsection 2.6 a circular design process with intermittent evaluations will be used to create a satisfactory product.

3 Product development

In this section we will give a broad understanding of the product development by discussing the different prototypes and the choices made in the process of making and improving prototypes.

Allard Roeterink (personal communication, 25 March, 2026) - a creative technologist and educator - advised to produce a first prototype with LEGO and gave advice on which principles could be used to make a working prototype, such as pulleys, weights and magnets. The subsequent two ideas for a product can be seen in Figure 6 and Figure 7.

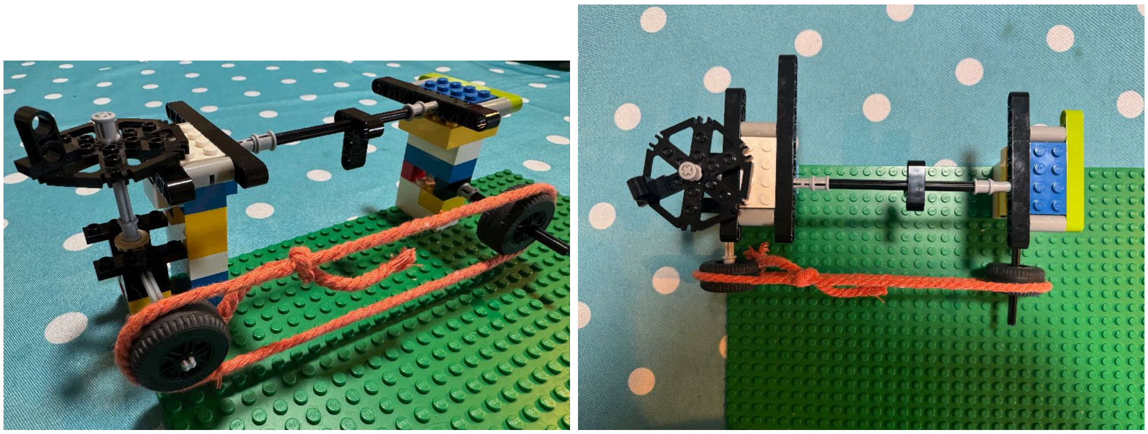


Figure 6: Here are two images depicting the first prototype with a pulley feedback system made with LEGO.

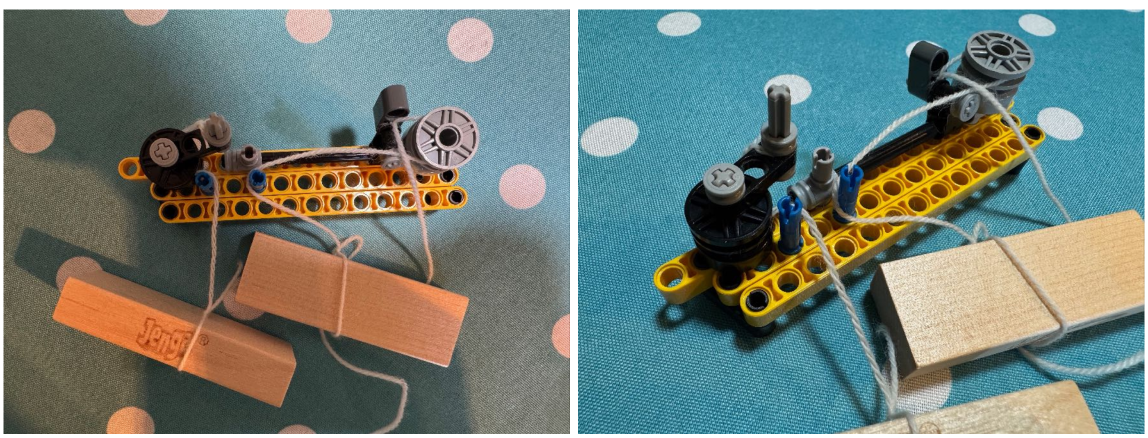


Figure 7: Here are two images depicting the first prototype with a weights feedback system made with LEGO.

As discussed in subsection 2.2 an action-based embodied design requires the posing of a motor problem with performance feedback. The motor problem which has to be solved is moving a point over a circle and moving a point on a horizontal line the same distance as the point on the circle has moved. This is similar to the digital design as mentioned in subsubsection 2.5.2.

The feedback system of the prototype shown in Figure 6 is given by the knot on the rope which indicates where the LEGO block on the rail should be positioned. The knot moves when the roundish gear is turned using the handle. However, this feedback shows the way a user is supposed to move. A user of an action-based embodied design is supposed to struggle to solve the motor problem which is not the case with this first prototype. When further developing this prototype the feedback system should be changed such that a user would struggle to solve the motor problem. The feedback system of the other prototype shown in Figure 7 is given by the height of the weights (JENGA blocks). Moving the handle on the circle or the rail moves the weights up or down. When the blocks are on the same height (if the prototype is aligned) then the same distance has been traveled on the circle and the rails.

The prototype with a pulley system has the advantage of versatility if an electrical circuit is introduced. The feedback mechanism would change slightly by introducing a magnet where the knot is and giving the handle on the rail a hall effect or reed switch which would close the electrical circuit once the magnet is close enough. The electrical circuit could contain a light, motor or small speaker which turns on as a feedback mechanism. A drawback of this design is that it can become more elaborate and be less lowtech than the prototype with weights. A lowtech design has as an advantage that it will probably be less expensive. The versatility of the prototype with a pulley system would mean that the product can be adjusted to a student's needs which would make it useful for students with and without a visual impairment. This adaptability of the design outweighs that it is more complicated.

Figure 8 shows the second prototype which is based on the LEGO prototype with a pulley feedback system.

Figure 8: Here are images which show the inside and outside of the second prototype.

The physical design should not be small, because then the movement to solve the motor problem would be small as well. However, the product is meant to be used in classes and thus should fit on a desk. A radius of five centimetres ($r = 5$ cm) would give a circumference of approximately 32 centimetres. This would mean the product would be at least 42 centimetres long and 10 centimetres wide. This is sufficiently big and might even be a little bit on the big side for on a desk. The prototype as can be seen in Figure 8 has a length of approximately 58 centimetres, width of 15 centimetres and depth of 4.4 centimetres. The length has been increased to ensure that the magnet on the pulley can reach the end of the horizontal line in the lid.

As can be seen in Figure 8 most of the moving parts such as the grips, wheels and gears have been made using 3D printers. 3D printing provides the opportunity to custumise the gears, such that they for example have a radius of five centimetres and can accomodate a ballchain. Moreover, if a school has a 3D printer available they could build most of the prototype themselves. This could make the prototype less expensive for schools with these resources.

For the conveyor belt to which the magnet is attached we choose to use a ballchain. Per metre a ballchain is cheaper than a rubber conveyor belt.

The electrical circuit in the second prototype is not completely soldered which can be seen in Figure 8. A breadboard is attached to the lid which connects the different components, namely the batteries, hall effect, LED and on/off switch. The breadboard makes it possible to add, change or replace components in the electrical circuit. For example the electrical circuit could be changed to incorporate a small motor. This would require some amplifiers due to how the hall effect works. In later prototypes the bread board might no longer be necessary and everything could be soldered together.

As mentioned above the motor problem is moving two points the same distance, one of the points on a circle and the other on a horizontal line. In Figure 9 different positions of the handles show how the second prototype would work. The LED at the top of the box turns on when both grips have travelled approximately the same distance from their starting point. If one of the grips is farther away than the other grip then the LED is off. This is achieved by the magnet which moves when the wheel underneath the grip in the circle turns and by the arm attached to the grip in the line with a hall effect attached to it. The hall effect closes the electrical circuit (if it is turned on with the switch in the lower right corner) which turns on the LED. Users of the prototype are encouraged to find out when the LED turns on for example by a student worksheet (see section 10).

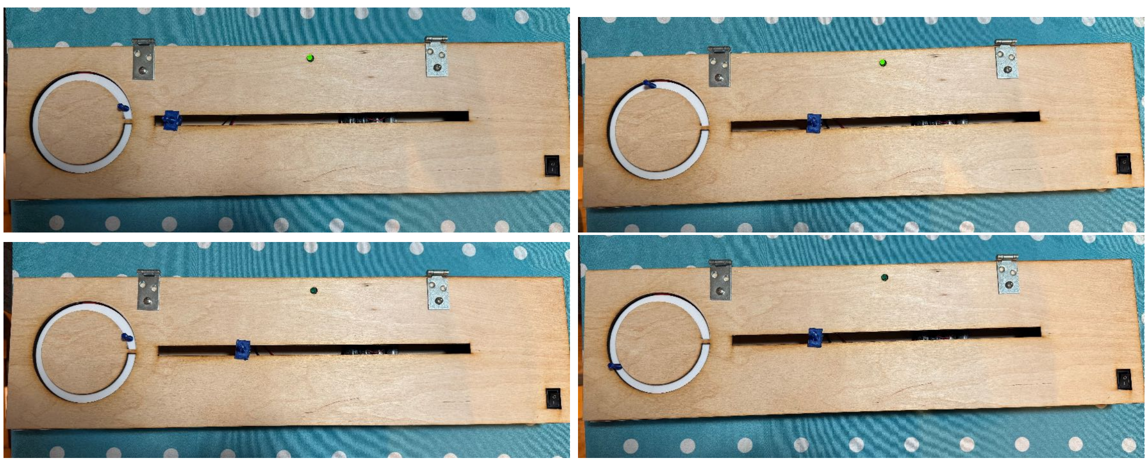


Figure 9: These are examples of the second prototype with different positions of the handles. In the upper two pictures the LED is on. In the lower two pictures the LED is off.

A part of this product is a short student worksheet (see section 10) and a short teacher guide (see section 9). The student worksheet can help students to engage with the prototype. The teacher guide briefly explains what the prototype is, how and when it could be used, and what the aim of using the prototype is. As discussed in subsection 2.2 a teacher guide and student worksheet help towards the product being easier to integrate into existing lessons. However, the connection to the lesson material still needs to be made by a teacher. This prototype is not very close to what most students learn, namely the sine graph as discussed in subsubsection 2.5.1. Therefore, it might be beneficial to make a more advanced prototype which contains all parts of the digital design as discussed in subsubsection 2.5.2 or three boxes each being one part of the before mentioned digital design.

An estimate of the costs for materials for this prototype is €30. This is excluding costs for maintaining and using machinery. A short list of approximate costs for different materials:

- Wooden plate for the box is €10.
- Filament for 3D printing parts is €3.
- A breadboard is €1.
- An on/off switch (big) is €1.
- A battery holder is €1.25.

- Two hinges are €3.
- Approximately six screws and two nuts are €1.
- A hall effect is €0.5.
- A magnet is €1.
- An LED is €0.1.
- An electrical resistance is €0.05.
- Approximately five metres of ballchain is €5.95.

For the prototype both the wooden plate and the ballchain were not completely utilized. The costs might lower when multiple prototypes are made and the wood and ballchain are utilized completely. However, this prototype will probably not cost less than €18 while using all the above mentioned materials. Considering one might need ten to fifteen of a similar prototype so it could be used in groups of students in a classroom, this is quite expensive.

Luttenberger et al., 2018 poses the suggestion that instructional strategies which enhance students' interest and motivation could reduce math anxiety as mentioned earlier in subsection 2.1. An artefact such as the second prototype could be such an instructional strategy for teachers. The lack of formal mathematics in the first encounter with this mathematical artefact might prompt students' curiousity instead of anxiety. The relative novelty of such an artefact can also invite curiousity. Moreover, this mathematical artefact makes the mathematical concept of the x-coordinate of the sine graph more tangible instead of introducing abstract notions of the unit circle and sine graph. Curiousity and tangibility together can lead to more familiarity with the concept after having connected the mathematical artefact to the formal mathematical concept.

3.1 Summary

Based on subsection 2.1 and subsection 2.2 choices were made with regard to the design of the prototype. Here we will briefly summarize which choices were made during the product development.

The prototype is meant to be an action-based embodied design as explained in subsection 2.2. Therefore, the prototype had to pose a motor problem and give (continuous) feedback. This prototype poses the problem of turning the LED on which happens when the two grips travel the same distance from their starting points. Here it is important that one of the grips does not move the other grip of the prototype. If this were the case, it would no longer be a (motor) problem to solve.

As set out in subsection 2.2 the product should be easy to integrate into existing classes. To work towards this aim the prototype is accompanied by a short student worksheet and a short teacher guide.

The product is meant to be affordable for schools. During the process of building the prototype choices were made with regards to materials, such as using a ballchain instead of a rubber conveyor belt. The approximate cost of materials needed to build this prototype is €30.

The product is meant to reduce math anxiety and foster conceptual understanding as mentioned in subsection 2.2. The product mainly does this through prompting curiousity and providing tangibility.

4 Evaluation

In this section we will evaluate whether the product fulfills the requirements which have been discussed in section 2 and we will discuss a pilot interview of the second prototype.

4.1 Product evaluation

In section 2 the requirements for the product are set out. In section 3 the prototype for the eventual product is discussed. Here we will discuss whether the prototype fulfills the requirements.

The most important criterium is that the prototype is a physical action-based embodied design with (continuous) feedback. As mentioned in section 3 the motor problem posed is moving two grips such that the LED turns on. The LED is the continuous feedback on the movements of the user of the prototype. In general the prototype fulfills the requirement of an action-based embodied design. Nonetheless, some improvements can be made with regards to accuracy and smoothly moving the grips.

Another criterium is that the product is affordable. As mentioned in section 3 the prototype costs approximately €30. In comparison to a tablet this is affordable. However, depending on how the product is implemented this can still be expensive. If the product is used in groups of two in a class of thirty students, then the school would need fifteen of these boxes for a class. Fifteen of these boxes would approximately cost €450. For a product which might be used just a couple of times a year this is rather expensive. Moreover, this price is without taking into account storing the boxes at a school, building them, or sending them with the post or letting someone of MATHerials deliver them. Taking all this into consideration this prototype is not yet affordable.

As discussed in subsection 2.2 it is important that the product is easy to integrate into the curriculum and classes. This poses an interesting question on how the prototype could be implemented in classes and how it can be connected to the curriculum. In the teacher guide (see section 9) for this prototype the assumption is made that every student experiences the product. However, this need not be the case. The prototype could be made bigger such that it could be used like a kind of demonstration tool. A demonstration tool has as a drawback that not all students will struggle with solving the motor problem the action-based embodied design poses which according to Alberto et al., 2022 impedes mathematical learning with such a design. However, a demonstration tool could still be used effectively in a classroom setting by letting students try to solve the motor problem together. The other question is how the prototype can be connected to the curriculum. In section 3 one way is mentioned, namely by building a prototype which incorporates all three parts of the digital action-based embodied design as set out in subsubsection 2.5.2. Especially the last part of the digital action-based embodied design is similar to what is usually taught about the sine graph in the Netherlands. Another way of connecting the prototype to the curriculum is by connecting it to the transformation of radials and degrees. This could be done by placing the grip in the circle at a certain angle to the starting point of the grip, and expressing this angle both in degrees and radials and connecting it to the distance travelled on the line as to turn on the LED. These are just two ideas about how to connect the prototype to the curriculum, but there might be more than these. It would be good to make this part of the product by elaborating on it in the teacher guide (section 9), adding more exercises to the student worksheet (section 10) and developping a workshop for teachers. To answer the above mentioned inquiries of how to implement the prototype in classes and how to connect it to the curriculum some pilot studies with teachers and/or students might give useful insights into for example the attitude of teachers towards new lesson material, what they need to implement it in class, their experience with implementing the product, how they perceive the reaction of their classes with respect to the product and the connection to the curriculum, and which implementation seems more effective according to both teachers and students.

Providing support for teachers with implementing the product and connecting it to the curriculum also supports teachers with facilitating conceptual understanding of mathematics among students. However, fostering conceptual understanding entails more than connecting the manipulative to the knowledge as set out in a textbook. Fostering conceptual understanding using a manipulative in class also requires some understanding of attentional anchors, the importance of gestures and translating the (physical) experience into a mathematical framework. This has not been added to the product. The development of a workshop for teachers on using manipulatives in classes can address these subjects from the perspective of recent research in mathematics education and give some examples of gesturing from experiments with manipulatives.

The product now consists of a prototype, a short student worksheet and a short teacher guide. As mentioned in subsection 2.4 one of the target audiences is (educational) designers. The product does not address this target audience yet. Parts of the teacher guide can be used by designers, such as what the goal for students is and how it can be implemented. However, a text or other medium more focused on designers as an audience will help create visibility for action-based embodied designs. A possible example would be a flyer which sets out the design principles behind the product.

4.2 Process evaluation

Here we will discuss the steps taken to evaluate the prototype, the knowledge gained from these evaluations and which steps could be taken to further evaluate the product. Some quotes are used in this text which have been translated from Dutch to English.

The second prototype which can be seen in Figure 8 and Figure 9 was evaluated with a pilot interview. In total two participants were individually interviewed, whom will be reffered to as participants A and B. During an interview a small introduction was given about the prototype after which the participant was asked to use the prototype to solve some questions, such as what is the (mathamatical) rule behind turning on the LED. The interview questions (in Dutch) can be found in section 8.

Both participants (almost) immediately started using the prototype.

Participant A: *"It is a bit fidly, isn't it?"*
Participant B: *"The grip in the line moves rather stiffly."*

As can be seen from the quotes, the participants had some difficulty with solving the motor problem due to the prototype being rather fidly, rough and the prototype did not have enough accuracy. The lack of accuracy sometimes led to losing the LED for a while as can be seen from the quotes below.

Participant A: *"I lost it."*
Participant B: *"Oops, now I lost it."*

Adressing this in the next prototype is important to improve the experience of users.

In general both participants could imagine themselves using the product in their courses/lessons. Participant A mentioned that *"It does make you curious, which is sometimes difficult to achieve with mathematics."* and Participant B said that *"Starting to use the prototype is very simple and the entry level is very low. Any pupil can try it and can get quite far, although explaining the idea behind it is another matter."* Each participant indicated a different way how they would use the product during a lesson. Participant A would prefer to use the product in a class discussion which would give more control over the interpretation and the product might be interpreted as being more special by pupils due to only one product being necessary in a class discussion. The other participant indicated that forming groups of two students would be preferable such that each student could interact with a product and some interaction between students would be created. After the groups have completed some exercises on their own, a class discussion would follow to connect the product to the mathematics curriculum.

Participant A: *"The vulnerability lies in the next step, connecting it to the mathematical concept."*

As examplified in the above mentioned quote, the participants expressed some concerns with regard to connecting the prototype to the curiculum. As has been discussed in subsection 4.1 the prototype can be connected to the curriculum in multiple ways. To make the product easier to integrate into lessons it is relevant to study and/or to conduct interviews into some of the different ways to connect the prototype to the curriculum. Such studies and interviews can give more insight into what is necessary to succesfully connect an action-based embodied design to existing lesson materials and curricula.

Participant B: *"Some teachers see this as playing. Then they think 'now we will do real mathematics' eventhough this is also real mathematics."*
Participant B: *"It helps teachers if you prepare everything for them in advance. For example, explaining*

the transition to a standard textbook would help due to teachers finding it difficult to teach outside of the textbook."

As examplified by the quotes above, another challenge is that teachers might not be acquainted with using manipulatives during lessons and connecting a manipulative to the curriculum. The teacher guide (see section 9) could help with this. However, a workshop or presentation about an action-based embodied design would make it possible to inform teachers better both about the design and about embodied cognition theory. Moreover, as mentioned in subsection 4.1 the teacher guide only mentions one way of implementing the prototype and does not specify how to connect the prototype to the curriculum. The teacher guide still needs some improvements, such as elaborating the different ways of how to connect the prototype to the curriculum.

Some improvements for the prototype mentioned by the participants are:

- making the lid a writable surface such that students can mark certain positions of the handles,
- making the handles/grips less slim,
- adding a little speaker for sound or motor for tactile feedback for pupils with a visual impairment,
- and adding a mechanism to close the lid such that the magnet and hall effect make better contact.

Both participants in this pilot interview were educators. With some improvements a new prototype could be tested with students of secondary schools to gain more insight in how students react to such a product and whether it improves their understanding and reduces their math anxiety. Moreover, it would be insightful to know what students would suggest as improvements for a prototype. This has also been mentioned by particiapant B, namely *"Eventually it is about how students use it, and they do sometimes come up with other things."*

5 Conclusion

The product is based on embodied cognition theory and could as a manipulative for learning reduce math anxiety and encourage conceptualisation through curiosity and tangibility. The design of the product is based on the action-based embodied design for the sine graph and specifically the part which concerns itself with the x-coordinate of the sine graph.

As explained in section 3 only one of the two first prototypes has been worked-out. For the future it might be interesting to work-out the other prototype (with the weights system). This would be interesting because both prototypes could be compared and studied to gain more insight in how people and students would learn the same concept with different manipulatives.

The prototype is a succesful translation of the digital action-based embodied design as set out in subsubsection 2.5.2 to a physical design. As mentioned in section 4, the user experience of the product still needs to be improved by making the prototype more accurate which can be achieved by better placing the magnet and the hall effect. Moreover, the student worksheet and teacher guide need to be improved to make connecting the prototype to the curriculum and implementing the prototype easier. Other improvements mentioned in section 4 are:

- making the lid a writable surface such that students can mark certain positions of the handles,
- making the handles/grips less slim,
- adding a little speaker for sound or motor for tactile feedback for pupils with a visual impairment,
- and adding a mechanism to close the lid such that the magnet and hall effect make better contact.

Some not yet mentioned improvements which could be made are:

- install a LEDstrip instead of a single LED,
- reducing the height of the outer box,
- adding a mechanism to keep the ballchain under tension,

- and using smaller hinges or attaching the hinges to the inside of the box such that they do not form a distraction and the design looks better.

Due to a mistake while building the prototype the height of the outer box is approximately four centimetres. The gears could be put closer together which would reduce the height of the mechanism and thus also the height of the outer box. Installing an LEDstrip instead of a single LED would change the focus from a point (the single LED) to a more general focus on the surface of the box. This could be interesting for eye-tracking studies which could reveal relevant data concerning attentional anchors for this design.

Eventhough the prototype is succesful as a translation of a digital design to a physical action-based embodied design, the product as a whole needs more research on different areas such as effectiveness and implementation in classes. Such research can lead to improvements such that the product may be more succesful in being used in an educational setting in schools. Moreover, an exploration of how the product could be used in other educational settings such as a museum might also be interesting.

Bibliography

Alberto, R., Shvarts, A., Drijvers, P., & Bakker, A. (2022). Action-based embodied design for mathematics learning: A decade of variations on a theme. *International Journal of Child-Computer Interaction*, *32*, 100419. https://doi.org/https://doi.org/10.1016/j.ijcci.2021.100419

Bakker, H., Boon, B., Bos, D., van Dijk, B., Doekes, W., Gietema, N., Greefkens, S., Peereboom, H., van den Reek, K., Reijenga, R., Schaberg, G., Sinkeldam, R., & Wallien, C. (2014). *Moderne wiskunde: 4 havo b* (11th ed.). Noordhoff Uitgevers Groningen.

Boonstra, K., Kool, M., Shvarts, A., & Drijvers, P. (2023). Theories and practical perspectives on fostering embodied abstraction in primary school geometry education. *Frontiers in Education*, *Volume 8*. https://doi.org/10.3389/feduc.2023.1162681

de Freitas, E., Ferrara, F., & Ferrari, G. (2018). The coordinated movements of collaborative mathematical tasks: The role of affect in transindividual sympathy. *ZDM*, *51*, 305–318. https://api.semanticscholar.org/CorpusID:149658003

Dubberly, H. (2004). *How do you design?* Dubberly Design Office.

Farina, M. (2020). Embodied cognition: Dimensions, domains and applications. *Adaptive Behavior*, *29*, 105971232091296. https://doi.org/10.1177/1059712320912963

Foley, A. E., Herts, J. B., Borgonovi, F., Guerriero, S., Levine, S. C., & Beilock, S. L. (2017). The math anxiety-performance link: A global phenomenon. *Current Directions in Psychological Science*, *26*(1), 52–58. https://doi.org/10.1177/0963721416672463

Gravemeijer, K., Bruin-Muurling, G., Kraemer, J., & van Stiphout, I. (2016). Shortcomings of mathematics education reform in the netherlands: A paradigm case? *Mathematical Thinking and Learning*, *18*(1), 25–44. https://doi.org/10.1080/10986065.2016.1107821

HU. (2026). *Lectoraat wiskundig en analytisch vermogen van professionals* [visited on 12-03-2026]. https://www.hu.nl/onderzoek/wiskundig-en-analytisch-vermogen-van-professionals#Onderzoekslijnen

Luttenberger, S., Hackl-Wimmer, S., & Paechter, M. (2018). Spotlight on math anxiety. *Psychology Research and Behavior Management*, *11*, 311–322. https://doi.org/10.2147/PRBM.S141421

Núñez, R., Edwards, L., & Matos, J. (1999). Embodied cognition as grounding for situatedness and context in mathematics education. *Educational Studies in Mathematics*, *39*, 45–65. https://doi.org/10.1023/A:1003759711966

Ramirez, G., Shaw, S., & Maloney, E. (2018). Math anxiety: Past research, promising interventions, and a new interpretation framework. *Educational Psychologist*, *53*, 1–20. https://doi.org/10.1080/00461520.2018.1447384

Shapiro, L., & Stolz, S. (2019). Embodied cognition and its significance for education. *Theory and Research in Education*, *17*, 19–39. https://doi.org/10.1177/1477878518822149

Shvarts, A., & van Helden, G. (2021). Embodied learning at a distance: From sensory-motor experience to constructing and understanding a sine graph. *Mathematical Thinking and Learning*, *25*, 1–28. https://doi.org/10.1080/10986065.2021.1983691

6 Appendix: AI statement

During the process of writing this report I have used Deepl/Google translate to translate Dutch sentences and terminology to English.

7 Appendix: Concept maps

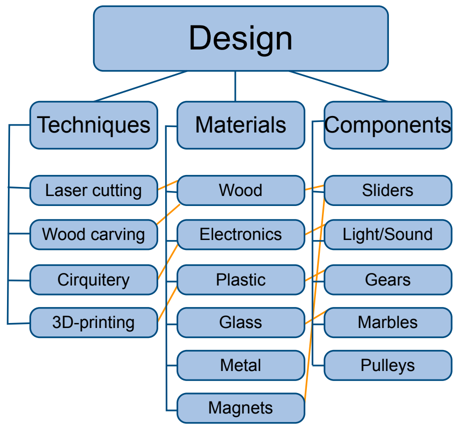


Figure 10: This is the concept map for a design.

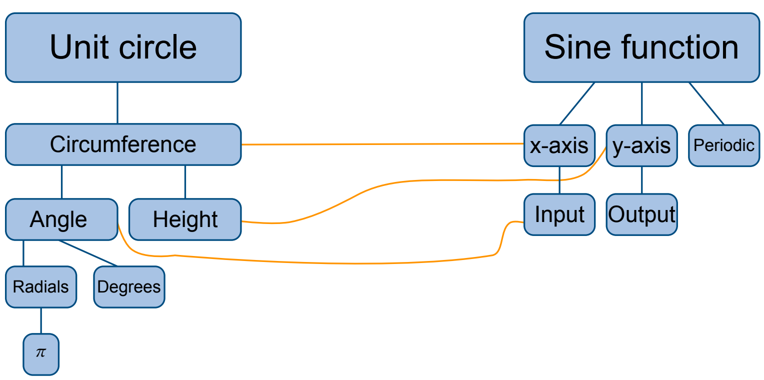


Figure 11: This is the concept map for the sine function.

8 Appendix: Interview questions for evaluation of a prototype

See the next page.

Interview leidraad

Voorbereiding

- Spullen ophalen
- Spullen klaarzetten, camera aan zijkant zetten
- Even video/geluid testen
- Printen werkblad
- Extra spullen meenemen, zoals pen & papier voor aantekeningen

Administratie

- Informatiebrief
- Toestemmingsformulier
- Camera aanzetten

Start

- Kennismaken (wie je bent, wat je doet)
- Doelen van dit onderzoek/pilot/interview:
  van digitaal naar concreet/fysiek ontwerp
  voor feedback
- Inleven in zowel leerlingen als docent

Introductie

Je ziet een doos met een aan/uit-knop, een (groen) ledje bovenaan, een greepje in een horizontale lijn en een greepje in een cirkelvormige lijn. Voordat je met de oefeningen begint, moet je de doos aanzetten met de aan/uit-knop. In de opdrachten zullen we het **greepje in de cirkelvormige lijn punt A** noemen en het **greepje in de horizontale lijn punt B**.

Je hebt twee punten die je kan bewegen. Deze hier (rond), en deze hier (recht). Het is jouw taak om de punten zo te bewegen dat je het ledje aangaat/aanblijft.

Opdrachten

Ontdekken

1. Probeer het groene ledje aan te laten gaan. (Mocht niks werken dan staat de doos uit of is de batterij op.)
2. Probeer nog een andere plek te vinden waar het groene ledje aangaat.
3. Probeer nu het ledje aan te houden terwijl je zowel punt A als punt B beweegt.

Reflectie op ontdekken

1. Is het gelukt om het ledje continu te laten branden bij de derde vraag van ontdekken?
2. Hoe doe je dit (het ledje aangaat)? Waar keek je naar om dit te bereiken?
3. Wat zou een goede regel zijn om het ledje te laten branden?

X-coördinaat oefening

1. Zet punt A ergens neer. Waar zou punt B nu moeten staan?
2. Waar moet punt B staan als punt A op een kwart van de cirkel staat ($\pi/4$)?
3. Waar moet punt B staan als punt A op een derde van de cirkel staat ($2\pi/3$)?

Eindvragen

- Wat vond je van je ervaring?
- Implementatie:
  Zie je jezelf dit doen met leerlingen?
  Zou het haalbaar zijn met leerlingen?
  Hoe zou je dit in de klas inzetten? (Groepjes, iedereen een box)
- Drempels:
  In hoeverre sluit dit aan bij lesstof (boek, etc.)?
  Zie jij andere drempels voor implementeren?
- Wat zouden docenten aan training nodig hebben om dit in te zetten in de klas?
- Welke dingen zou je veranderen/verbeteren?
- Hoe dicht/ver staat dit af van je huidige wiskunde activiteiten/didactiek?
- Voor welk onderwerp zou je zo'n tool willen hebben?

Afsluiting

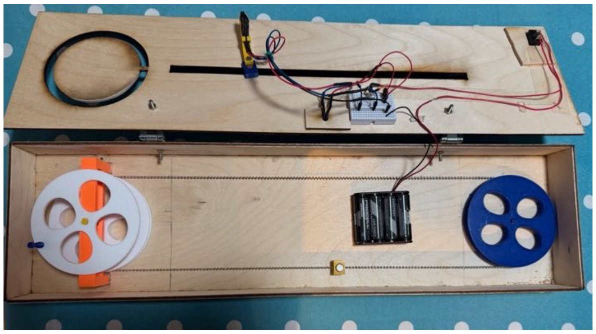
- Bedankt voor deelname
- Terugkoppeling: verslag doorsturen

9 Appendix: Short teacher guide

See the next page.

Korte docentenhandleiding

Voor het gebruik van het design voor de x-coördinaat van de sinusgrafiek

Het design dat u gaat gebruiken is vergelijkbaar met het design dat hierboven te zien is. Als het deksel dicht is, ziet u een doos met een aan/uitknop, een (groen) ledje bovenaan, een greepje in een horizontale lijn en een greepje in een cirkelvormige lijn.

Binnen in de doos zijn de elementen aanwezig die ook in de afbeelding hierboven te zien zijn. Alvorens het gebruik van dit design, is het goed om even te testen of de batterijen vervangen moeten worden en of de magneet op de juiste plek staat (de kleine gele kubus op de onderste draad in de afbeelding). De magneet hoort aan het begin van de rechte sleuf op de ketting te zitten wanneer het greepje in de cirkel boven tegen het houten lijntje staat.

Doel voor leerlingen

Het doel voor de leerlingen zal zijn om beide greepjes te bewegen zodat het ledje aangaat. Het ledje zal enkel aangaan wanneer het greepje in de cirkel ongeveer dezelfde afstand heeft afgelegd (gezien vanaf de bovenkant van het houten lijntje in de cirkel) als het greepje in de horizontale lijn (gezien vanaf de linkerkant van de horizontale lijn).

Wanneer te gebruiken

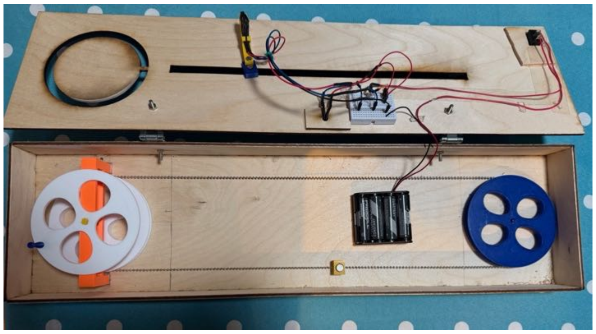
Het design is bedoeld om leerlingen een intuïtie te geven voor wat de x-coördinaat van de sinusgrafiek is door te bewegen. Hierom raden wij aan om dit design in te zetten net voordat de sinus functie/grafiek wordt geïntroduceerd.

Hoe te gebruiken

Met behulp van het korte werkblad voor leerlingen zullen leerlingen proberen de regel achter wanneer het ledje aangaat te ontdekken. Hiervoor is het aangeraden om leerlingen in kleine groepjes te laten werken met dit design. Nadat de leerlingen een tijdje met het design gewerkt hebben, kan deze ervaring verbonden worden met hoeken en de sinusgrafiek. Dit kan op allerlei manieren aangepakt worden. Een voorbeeld van een aanpak zou zijn om klassikaal te vragen aan leerlingen wat ze dachten dat de regel erachter is en hier verder op in te gaan. Uiteindelijk kunt u het antwoord geven en dit op een wiskundige manier met hoeken en de sinusgrafiek verder uitwerken.

Short teacher guide

For the usage of the design for the x-coordinate of the sine graph

The design you will be using is similar to the one shown above. When the lid is closed, you will see a box with an on/off button, a (green) LED at the top, a handle in a horizontal line and a handle in a circular line.

Inside the box are the components as shown in the image above. Before using this design, it is advisable to check whether the batteries need replacing and whether the magnet is in the correct position (the small yellow cube on the bottom wire in the image). The magnet should be at the start of the straight slot on the chain when the handle in the circle is aligned at the top of the wooden line.

Aim for pupils

The aim for the pupils will be to move both handles so that the LED lights up. The LED will only light up when the handle in the circle has travelled approximately the same distance (measured from the top of the wooden line in the circle) as the handle on the horizontal line (measured from the left-hand side of the horizontal line).

When to use

The design is intended to give pupils an intuitive understanding of the x-coordinate of the sine graph by moving. For this reason, we recommend using this design just before the sine function/graph is introduced.

How to use

Using the short worksheet for pupils, pupils will try to discover the rule governing this design. To do this, it is recommended that pupils work on this design in small groups. Once the pupils have spent some time working on the design, this experience can be linked to angles and the sine graph. There are many ways to approach this. One approach would be to ask the class what they think the underlying rule is and then explore this further. Finally, you can provide the answer and work it out in mathematical terms using angles and the sine graph.

10 Appendix: Short student worksheet

See the next page.

Kort leerlingen werkblad

Je ziet een doos met een aan/uitknop, een (groen) ledje bovenaan, een greepje in een horizontale lijn en een greepje in een cirkelvormige lijn. Voordat je met de oefeningen begint, moet je de doos aanzetten met de aan/uitknop. In de opdrachten zullen we het **greepje in de cirkelvormige lijn punt A** noemen en het **greepje in de horizontale lijn punt B**.

Het is jouw/jullie taak om punten A en B zo te bewegen dat het ledje aangaat/aanblijft.

Ontdekken

1. Probeer het ledje aan te laten gaan.
2. Probeer nog een andere plek te vinden waar het ledje aangaat.
3. Probeer nu het ledje aan te houden terwijl je zowel punt A als punt B beweegt.

Reflectie

1. Is het gelukt om het ledje continu te laten branden bij de derde vraag van ontdekken?
2. Hoe doe je dit (het ledje aanhouden)? Waar keek je naar om dit te bereiken?
3. Wat zou een goede regel zijn om het ledje te laten branden?

Extra oefeningen

1. Zet punt A ergens neer. Waar zou punt B nu moeten staan?
2. Waar moet punt B staan als punt A op een kwart van de cirkel staat?
3. Waar moet punt B staan als punt A op een derde van de cirkel staat?

Short student worksheet

You will see a box with an on/off switch, a (green) LED at the top, a handle on a horizontal line and a handle on a circular line. Before you start the exercises, you must switch the box on using the on/off switch. In the exercises, we will refer to **the handle in the circular line as point A** and **the handle in the horizontal line as point B**.

It is your task to move points A and B in such a way that the LED lights up or stays lit.

Discover

1. Try to get the LED to light up.
2. Try finding another spot where the LED lights up.
3. Now try to keep the LED lit whilst moving both point A and point B.

Reflection

1. Did you manage to keep the LED lit continuously during the third question in the 'Discover' section?
2. How do you do this (keep the LED lit)? What did you look at to achieve this?
3. What would be a good rule to keep the LED lit?

Extra exercises

1. Place point A somewhere. Where should point B be now?
2. Where should point B be if point A is at a quarter of the way round the circle?
3. Where should point B be if point A is at a third of the way round the circle?

11 Appendix: Quotes from pilot study in Dutch

11.1 Participant A

"Het maakt nieuwsgierig. Het is echt zo'n mystery box."
"Beetje fidly he."
"Gelijkmatige beweging hier (cirkel) en dan een gelijkmatige beweging hier (lijn)."
"Ben hem kwijt."
"Helpt als je iets meer tolerantie hebt."
"Ik kan me ook wel voorstellen dat leerlingen dan denken en wat nu."
"Je kijkt naar het lampje, want daar komt de feedback natuurlijk vandaan."
"De kwetsbaarheid zit in de volgende stap, het verbinden met het wiskundige concept."
"Ik merk ook dat ik de behoefte krijg om markeringen te gaan maken."
"De kracht zit hem wel in het zelf ervaren, dus het is niet een demonstratie tool vind ik."
"Als je iets de klas inbrengt, iets tastbaars, dan is zo'n werkblad al niet meer zo interessant, dan ga je dat niet meer lezen."
"Het zet je wel in een nieuwsgierige stand, en dat is soms bij wiskunde moeilijk voor elkaar te krijgen."

11.2 Participant B

"Het schiet opeens een beetje door of zo."
"Ooh nou ben ik hem kwijt."
"Hij gaat ook heel stroef (lijn)."
"Het zou fijn zijn als je mee kon schuiven echt."
"Voor mijn braille leerlingen zou ik het op geluid willen."
"Leerlingen die wat meer kennis hebben zich op een gegeven moment gaan afvragen, ja maar het was toch -1."
"Ik denk zeker dat als je geen voorkennis hebt, dan gaat het heel soepel."
"Je krijgt een bepaald gevoel voor de beweging."
"Het is wel iets dat je graag wil, dat dat lampje aan blijft."
"Het is bijna als een spel of zo."
"Om te beginnen is heel simpel, de drempel is heel laag, elke leerling kan dit wel proberen en wel een eindje komen al is het verklaren wat anders."
"Dat doet materiaal volgens mij, dat leerlingen meer gemotiveerd raken om voor elkaar te krijgen en uit te zoeken. Ik denk dat het heel stimulerend werkt."
"Je gunt elke leerling die ervaring, toch?"
"Ik vraag me af hoe lang ze hiermee bezig zijn. 10 minuten, een kwartier."
"Als je alles kant en klaar voor ze (docenten) maakt, dan helpt dat wel. Ook bijvoorbeeld de overstap naar het gewone boek beschrijft. Dat vinden ze lastig om dingen buiten het boek te doen."
"Sommige docenten zien het als spelen en denken dan gaan we nu echte wiskunde doen (zoals in het boek) terwijl dit (materialen) ook echte wiskunde is."
"Het motiveert enorm, niet voor alle leerlingen hoor."
"Docenten trainen zal wel echt heel goed helpen. Ook door ze te helpen hier later op terug te komen, zodat de ervaring weer opgehaald wordt. En daar moet je docenten denk ik wel bij helpen en ook met goede opdrachten."
"Dit is gewoon wiskunde doen he, en ik denk dat dat gewoon heel erg helpt om je inzicht te krijgen."
"Je let vooral op dat lampje en je let nu echt op je beweging."
"Uiteindelijk gaat het erom hoe leerlingen ermee omgaan, en die komen soms met andere dingen weer hoor."